

514-Custom PCB Design

1) All the electronic components that will be soldered to display PCB

Main Control & Interface:

- **Microcontroller:** Seeed Studio XIAO ESP32-C3 (U1) – Soldered via header pins
- **Stepper Motor:** X27.168 Stepper Motor (4-pin) – Soldered via through-holes
- **Battery Connector:** JST-PH 2-pin Connector (2.0mm pitch, right-angle) (J2)
- **User Button:** 6×6mm Tactile Switch (SW1)

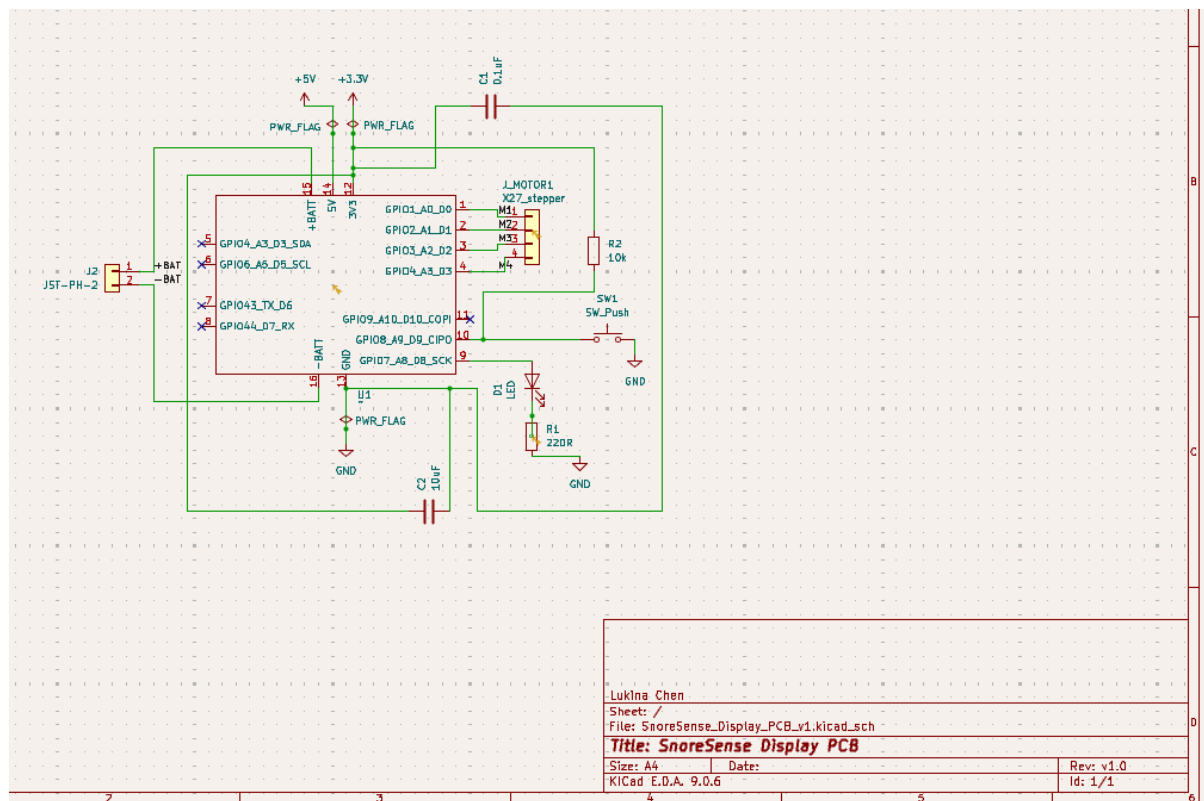
Resistors & LED:

- **Resistor (R1):** 220Ω – LED current limiting
- **Resistor (R2):** 10kΩ – Button pull-up resistor
- **LED:** 5mm LED (Through-hole) (D1)

Capacitors:

- **Capacitor (C1):** 0.1μF Ceramic – Decoupling
- **Capacitor (C2):** 10μF Ceramic – Bulk capacitance for 3.3V rail

2) A screenshot of circuit in the Schematic Editor window of KiCad



3) A screenshot of circuit board in the PCB Editor window of KiCad

